

Lablet 4 Solution

Instructions

Update configurations by performing the following tasks:

Task 1: Network Access

- Create VLANs on **Sw5** using data from the table.
- Configure **Sw5** interface e0/1 as an access port in the *Branch* VLAN
- Configure **Sw5** interfaces e1/0-3 for:
 - Data VLAN - *Desktops*
 - Voice VLAN - *VoIP*

Task 2: Dynamic Routing

Note: R2 is correctly pre-configured for OSPF and this task.

- Enable IPv4 OSPFv2 on **R1** and **Sw5** using process id 1.
- Ensure each OSPF router uses its *Loopback0* address as the Router ID.
- Configure OSPF routers to advertise all connected networks into area 0.
 - *Note: Connected networks includes all physical, VLAN, and Loopback interfaces with IP addresses configured.*
- Ensure only **R1** and **R2** can ever be the designated router.
 - **R1** must have the best chance of becoming the designated router.
- Configure **Sw5** so that it only attempts building OSPF adjacencies on interface e0/0

Task 3: Layer 2 Security

- Enable DHCP snooping on **Sw5** for the *Branch* and *Desktops* VLANs
- Ensure **Sw5** drops traffic from DHCP clients if the source MAC address doesn't match the snooped MAC address

Task 4: DHCP

Note: R1 is pre-configured as a DHCP server. Do NOT change any configuration on R1 for this task.

- Configure **Sw5** to forward DHCP requests from the *Branch* and *Desktops* VLANs to **R1** Loopback1 address.
- Configure **Branch-7** to use DHCP to assign an IPv4 address on interface e0/0.
- Ensure **Branch-7** uses the default route provided by DHCP.

Task 1: Network Access

Sw5

```
enable
config t
!
vlan 10
  name Desktops
  exit
vlan 20
  name VoIP
  exit
vlan 50
  name Branch
  exit
!
inter e0/1
  no shut
  switchport mode access
  switchport access vlan 50
  exit
!
interface range e1/0-3
  no shut
  switchport mode access
  switchport access vlan 10
  switchport voice vlan 20
!
end
```

Verifications

Sw5

```
show vlan b
```

VLAN	Name	Status	Ports
1	default	active	Et0/2, Et0/3

```

10 Desktops active Et1/0, Et1/1, Et1/2, Et1/3
20 VoIP active Et1/0, Et1/1, Et1/2, Et1/3
50 Branch active Et0/1
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup
1005 trnet-default act/unsup

```

```
show interfaces ether0/1 switchport | inc Name|Access Mode VLAN|Voice VLAN
```

```

Name: Et0/1
Access Mode VLAN: 50 (Branch)
Voice VLAN: none

```

```
show interfaces ether1/0 switchport | inc Name|Access Mode VLAN|Voice VLAN
```

```

Name: Et1/0
Access Mode VLAN: 10 (Desktops)
Voice VLAN: 20 (VoIP)

```

```
show interfaces ether1/1 switchport | inc Name|Access Mode VLAN|Voice VLAN
```

```

Name: Et1/1
Access Mode VLAN: 10 (Desktops)
Voice VLAN: 20 (VoIP)

```

```
show interfaces ether1/2 switchport | inc Name|Access Mode VLAN|Voice VLAN
```

```

Name: Et1/2
Access Mode VLAN: 10 (Desktops)
Voice VLAN: 20 (VoIP)

```

Task 2: Dynamic Routing

R1

```

enable
config t
!
interface e0/0
 ip ospf priority 100
 exit
!
router ospf 1
 router-id 192.168.0.1
 network 10.1.0.0 0.0.255.255 area 0
 network 192.168.0.0 0.0.255.255 area 0
 network 10.4.1.0 0.0.0.255 area 0
!
end

```

Sw5

```

enable
config t
!
interface e0/0
 ip ospf priority 0
 exit
!
router ospf 1
 router-id 192.168.0.5
 passive-interface default
 no passive-interface Ethernet0/0
 network 10.5.0.0 0.0.255.255 area 0
 network 192.168.0.0 0.0.255.255 area 0
 network 10.4.1.0 0.0.0.255 area 0
!
end

```

Verifications

R1

```
show ip ospf inter brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Lo0	1	0	192.168.0.1/32	1	LOOP	0/0	
Lo1	1	0	192.168.10.1/32	1	LOOP	0/0	
Et0/0	1	0	10.4.1.1/24	10	BDR	1/1	
Et0/3	1	0	10.1.103.1/24	10	DR	0/0	
Et0/2	1	0	10.1.102.1/24	10	DR	0/0	
Et0/1	1	0	10.1.101.1/24	10	DR	0/0	

```
show ip ospf interface e0/0 | inc Prio
```

```
! Priority must be higher than 50 to be better than R2
Transmit Delay is 1 sec, State BDR, Priority 100
```

```
show ip proto | inc ospf|Router
```

```
Routing Protocol is "ospf 1"
Router ID 192.168.0.1
```

Sw5

```
show ip ospf inter brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Lo0	1	0	192.168.0.5/32	1	LOOP	0/0	
Lo1	1	0	192.168.10.5/32	1	LOOP	0/0	
E0/0	1	0	10.4.1.5/24	10	DROTH	2/2	
Vl50	1	0	10.5.50.1/24	1	WAIT	0/0	
Vl20	1	0	10.5.20.1/24	1	WAIT	0/0	
Vl10	1	0	10.5.10.1/24	1	WAIT	0/0	

```
show ip ospf interface e0/0 | inc Prio
```

```
Transmit Delay is 1 sec, State DROTHER, Priority 0
```

```
show ip proto | inc ospf|Router
```

```
Routing Protocol is "ospf 1"
Router ID 192.168.0.5
```

```
show ip protocols | beg Passive
```

```
Passive Interface(s):
Loopback0
Loopback1
Vlan10
Vlan20
Vlan50
Routing Information Sources:
Gateway      Distance    Last Update
192.168.0.1      110        00:00:49
192.168.0.2      110        00:00:49
Distance: (default is 110)
```

Task 3: Layer 2 Security

Sw5

```
enable
config t
!
ip dhcp snooping
ip dhcp snooping vlan 10
ip dhcp snooping vlan 50
ip dhcp snooping verify mac-address
end
```

Verifications

Sw5

```
sho ip dhcp snooping
```

```
Switch DHCP snooping is enabled
Switch DHCP gleaning is disabled
DHCP snooping is configured on following VLANs:
10,50
DHCP snooping is operational on following VLANs:
10,50
Proxy bridge is configured on following VLANs:
none
Proxy bridge is operational on following VLANs:
none
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is enabled
circuit-id default format: vlan-mod-port
remote-id: aabb.cc00.3d00 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:
```

Interface	Trusted	Allow option	Rate limit (pps)
-----	-----	-----	-----

Note: This command shows the snooping database. Requires Task 4 to have been configured correctly, and that Branch-7 and Desk-8 clients release/renew their leases after snooping is enabled.

```
show ip dhcp snooping binding
```

MacAddress	IpAddress	Lease(sec)	Type	VLAN	Interface
AA:BB:CC:00:3C:00	10.5.10.12	36532	dhcp-snooping	10	Ethernet1/0
AA:BB:CC:00:3A:00	10.5.50.12	36396	dhcp-snooping	50	Ethernet0/1
Total number of bindings: 2					

Task 4: DHCP

Sw5

```
enable
config t
!
interface vlan 10
 ip helper-address 192.168.10.1
 exit
interface vlan 50
 ip helper-address 192.168.10.1
 exit
!
end
```

Branch-7

```
enable
config t
!
interface e0/0
 ip address dhcp
 exit
!
end
```

Verifications

Sw5

```
show ip inter Vlan10 | inc Vlan|Helper

Vlan10 is up, line protocol is up
  Helper address is 192.168.10.1

show ip inter Vlan50 | inc Vlan|Helper

Vlan50 is up, line protocol is up
  Helper address is 192.168.10.1
```

R5

Note: These checks should work independent of other tasks to check configuration.

```
show ip interf e0/0 | inc DHCP

  Address determined by DHCP

show run | inc ip route 0.0.0.0

! No output returned. There should NOT be a static default route configured
```

Note: These checks only work if Task 1 is completed correctly.

```
show logging | inc DHCP

*Jul 28 16:39:57.181: %DHCP-6-ADDRESS_ASSIGN: Interface Ethernet0/0 assigned DHCP address 10.5.50.11, mask 255.255.255.0, hostname Branch-7

show ip route 0.0.0.0

Routing entry for 0.0.0.0/0, supernet
  Known via "static", distance 254, metric 0, candidate default path
  Routing Descriptor Blocks:
  * 10.5.50.1
    Route metric is 0, traffic share count is 1
```